

Hints & Solutions

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Q1 (1) - Chord, Focal Chord and Its Properties.

Given equation of directrix, $x = 0$, $e = \frac{1}{2}$ and focus $(3, 0)$.

Equation of ellipse will be

$$\sqrt{(x-3)^2 + y^2} = \frac{1}{2}|x|$$

$$\Rightarrow (x-3)^2 + y^2 = \frac{x^2}{4}$$

$$\Rightarrow 4(x-3)^2 + 4y^2 = x^2$$

$$\Rightarrow 3x^2 - 24x + 4y^2 + 36 = 0$$

$$\Rightarrow 3(x-4)^2 + 4y^2 = 12$$

$$\Rightarrow \frac{(x-4)^2}{4} + \frac{y^2}{3} = 1$$

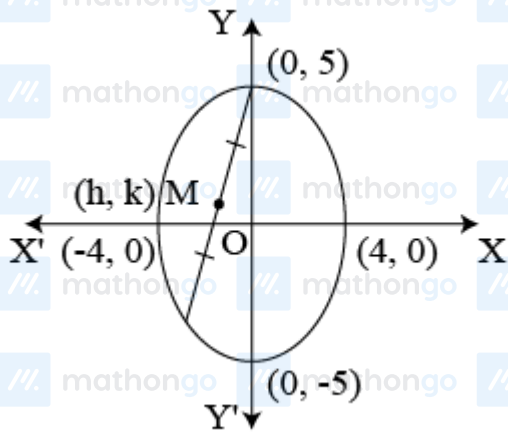
$$\therefore a = 2, b = \sqrt{3}, \text{ centre } (4, 0)$$

$$\therefore \text{Equation of auxiliary circle is } (x-4)^2 + y^2 = 4$$

$$\text{i.e. } x^2 + y^2 - 8x + 12 = 0$$

Q2 (3) - Chord, Focal Chord and Its Properties.

We have, $4 \cos \theta = 2h$ and $5(1 + \sin \theta) = 2k$



We know that,

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\Rightarrow \frac{h^2}{4} + \left(\frac{2k}{5} - 1\right)^2 = 1$$

$$\Rightarrow \frac{h^2}{4} + \frac{\left(k - \frac{5}{2}\right)^2}{\frac{25}{4}} = 1$$

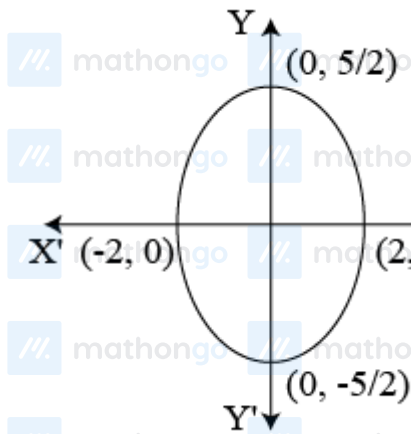
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$$\Rightarrow \frac{x^2}{4} + \frac{\left(y - \frac{5}{2}\right)^2}{\frac{25}{4}} = 1 \dots (i)$$

Put $X = x$ and $Y = y - \frac{5}{2}$ in Eq. (i), we get

$$\frac{X^2}{4} + \frac{Y^2}{\frac{25}{4}} = 1$$



$$\therefore e = \sqrt{1 - \frac{4}{25} \times 4} = \frac{3}{5}$$

$$\therefore F_1 = \left(0, \frac{3}{2}\right), F_2 = \left(0, -\frac{3}{2}\right) \text{ in } XY.$$

Hence, in xy foci are $(0, 4)$ and $(0, 1)$

Q3 (1) - Chord, Focal Chord and Its Properties.

The coordinates of any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, whose eccentric angle is θ are $(a \cos \theta, b \sin \theta)$.

The coordinates of end points of latusrectum are $\left(ae, \pm \frac{b^2}{a}\right)$

$$\therefore a \cos \theta = ae$$

$$\text{and } b \sin \theta = \pm \frac{b^2}{a}$$

$$\Rightarrow \tan \theta = \pm \frac{b}{ae}$$

$$\therefore \theta = \tan^{-1} \left(\pm \frac{b}{ae} \right)$$

Q4 (3) - Chord, Focal Chord and Its Properties.

The coordinates of end points of the focal chord are $(a \cos \alpha, b \sin \alpha)$ and $(a \cos \beta, b \sin \beta)$. Therefore, the equation

by the focal chord is

$$\frac{x}{a} \cos \left(\frac{\alpha + \beta}{2} \right) + \frac{y}{b} \sin \left(\frac{\alpha + \beta}{2} \right) = \cos \left(\frac{\alpha - \beta}{2} \right)$$

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This passes through $(ae, 0)$

$$\therefore \frac{ae}{a} \cos \frac{\alpha+\beta}{2} = \cos \left(\frac{\alpha-\beta}{2} \right)$$

$$\Rightarrow \frac{\cos \left(\frac{\alpha+\beta}{2} \right)}{\cos \left(\frac{\alpha-\beta}{2} \right)} = \frac{1}{e}$$

$$\Rightarrow \frac{\cos \left(\frac{\alpha+\beta}{2} \right) - \cos \left(\frac{\alpha-\beta}{2} \right)}{\cos \left(\frac{\alpha+\beta}{2} \right) + \cos \left(\frac{\alpha-\beta}{2} \right)} = \frac{1-e}{1+e}$$

$$\Rightarrow \frac{2 \sin \frac{\alpha}{2} \sin \frac{\beta}{2}}{2 \cos \frac{\alpha}{2} \cos \frac{\beta}{2}} = \frac{1-e}{1+e}$$

$$\Rightarrow -\tan \frac{\alpha}{2} \tan \frac{\beta}{2} = \frac{1-e}{1+e}$$

$$\Rightarrow \tan \frac{\alpha}{2} \tan \frac{\beta}{2} = \frac{e-1}{e+1}$$

Q5 (2) - Chord, Focal Chord and Its Properties.

Let PQ and RS be two chords of the ellipse such that the eccentric angles of P, Q, R and S are α, β, γ and δ respectively. Suppose, PQ and RS meet major axis at $(C, 0)$ and $(-C, 0)$ respectively. Then,

$$\tan \frac{\alpha}{2} + \tan \frac{\beta}{2} = \frac{c-a}{c+a}$$

$$\text{and } \tan \frac{\gamma}{2} \tan \frac{\delta}{2} = \frac{-c-a}{-c+a} = \frac{c+a}{c-a}$$

$$\therefore \tan \frac{\alpha}{2} \tan \frac{\beta}{2} \tan \frac{\gamma}{2} \tan \frac{\delta}{2} = \frac{c-a}{c+a} \times \frac{c+a}{c-a} = 1$$

Q6 (3) - Tangent and its properties

$$\text{The ellipse is } \frac{x^2}{4} + \frac{y^2}{3} = 1$$

$$\therefore a^2 = 4 \text{ and } b^2 = 3$$

Slope of the line $y + 2x = 4$ is -2

Any tangent to the ellipse is

$$y = mx \pm \sqrt{a^2 m^2 + b^2}$$

Given, it is perpendicular to given line, then $m = 1/2$.

$$\therefore y = \frac{1}{2}x \pm 2$$

$$\therefore 2y = x \pm 4$$

Q7 (4) - Tangent and its properties

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Equation of line $x + 2y - 4 = 0$... (i)

Equation of tangent at any point is

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

$$\Rightarrow \frac{x}{2} \cos \theta + \frac{y}{\sqrt{3}} \sin \theta = 1 \dots (ii)$$

On comparing Eqs. (i) and (ii), we get

$$\cos \theta = \frac{1}{2} \text{ and } \sin \theta = \frac{\sqrt{3}}{2}$$

The point of contact $(a \cos \theta, b \sin \theta)$ is $\left(1, \frac{3}{2}\right)$.

Q8 (3) - Tangent and its properties

Any tangent to the ellipse is

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

On putting $y = 0$ then $x = 0$, we get the points A and B , where it meets the axes as

$$A \left(\frac{a}{\cos \theta}, 0 \right), B \left(0, \frac{b}{\sin \theta} \right)$$

If (h, k) be mid-point of AB , then

$$2h = \frac{a}{\cos \theta}$$

$$2k = \frac{b}{\sin \theta}$$

$$\therefore \cos \theta = \frac{a}{2h}$$

$$\sin \theta = \frac{b}{2k}$$

Since, $\cos^2 \theta + \sin^2 \theta = 1$

$$\therefore \frac{a^2}{4h^2} + \frac{b^2}{4k^2} = 1$$

$$\Rightarrow a^2 y^2 + b^2 x^2 = 4x^2 y^2$$

Q9 (2) - Tangent and its properties

Let (h, k) be the foot of perpendicular drawn from the focus $S(ae, 0)$ on tangent

$$y = mx + \sqrt{a^2 m^2 + b^2} \dots (i)$$

at point P on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

$$\text{Then, } k = mh + \sqrt{a^2 m^2 + b^2}$$

$$\Rightarrow (k - mh)^2 = a^2 m^2 + b^2 \dots (ii)$$

Also, SP is perpendicular to Eq. (i).

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$$\therefore m \times \frac{k-0}{h-ae} = -1$$

$$\Rightarrow h + mk = ae \dots (iii)$$

From Eqs. (ii) and (iii), we get

$$\begin{aligned} (k - mh)^2 + (h + mk)^2 &= a^2 m^2 + b^2 + a^2 e^2 \\ \Rightarrow (h^2 + k^2) (1 + m^2) &= a^2 (1 + m^2) \\ \Rightarrow h^2 + k^2 &= a^2 \end{aligned}$$

Hence, the locus of (h, k) is $x^2 + y^2 = a^2$.

Q10 (4) - Tangent and its properties

Ellipse is $\frac{x^2}{16} + \frac{y^2}{4} = 1$; $a^2 = 16$, $b^2 = 4$

Let the equation of circle with centre $C(1, 0)$ be $(x - 1)^2 + y^2 = r^2$.

Since, the circle is to be largest, so it will touch the ellipse at some point $P(a \cos \theta, b \sin \theta)$ where tangent is

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

$$\text{whose slope of } m_1 = -\frac{b \cos \theta}{a \sin \theta} = -\frac{1}{2} \frac{\cos \theta}{\sin \theta} \dots (i)$$

$$\begin{aligned} \text{Also, } m_2 &= \text{Slope of } CP \\ &= \frac{b \sin \theta}{(a \cos \theta - 1)} = \frac{2 \sin \theta}{4 \cos \theta - 1} \dots (ii) \end{aligned}$$

$$\text{But } m_1 m_2 = -1$$

$$\therefore -\frac{1}{2} \frac{\sin \theta}{\cos \theta} \times \frac{2 \sin \theta}{4 \cos \theta - 1} = -1$$

$$\Rightarrow \cos \theta = 1/2$$

$$\therefore \text{Radius of circle is } \{(a \cos \theta - 1)^2 + b^2 \sin^2 \theta\}^{1/2}$$

$$\text{Or } \left\{ \left(\frac{4}{3} - 1 \right)^2 + 4 \left(1 - \frac{1}{a} \right)^2 \right\}^{1/2} = \sqrt{\frac{33}{9}} = \sqrt{\frac{11}{3}}$$

Q11 (3) - Tangent and its properties

We have,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$$\text{and } x^2 - y^2 = c^2$$

$$\Rightarrow \frac{dy}{dx} = x/y$$

The two curves will cut at right angles if

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$$\left(\frac{dy}{dx}\right)_{c_1} \times \left(\frac{dy}{dx}\right)_2 = -1$$

$$\Rightarrow \left(-\frac{b^2 x}{a^2 y}\right) \left(\frac{x}{y}\right) = -1$$

$$\Rightarrow \frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{1}{2} \left[\text{using } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \right]$$

On substituting these values in $x^2 - y^2 = c^2$, we get

$$\frac{a^2}{2} - \frac{b^2}{2} = c^2$$

$$\therefore a^2 - b^2 = 2c^2$$

Q12 (2) - Tangent and its properties

Let $y = mx + c$ be a common tangent to $9x^2 - 16y^2 = 144$ and $x^2 + y^2 = 9$

Since, $y = mx + c$ is a tangent to $9x^2 - 16y^2 = 144$

$$\therefore c^2 = a^2 m^2 - b^2$$

$$\Rightarrow c^2 = 16m^2 - 9 \quad \dots (i)$$

Now, $y = mx + c$ is a tangent to $x^2 + y^2 = 9$

$$\therefore \frac{c}{\sqrt{m^2+1}} = 3$$

$$\Rightarrow c^2 = 9(m^2 + 11) \dots (ii)$$

From Eqs. (i) and (ii), we get

$$16m^2 - 9 = 9 + 9m^2$$

$$m = \pm 3\sqrt{\frac{2}{7}}$$

Putting the value of m in Eq. (ii), we get

$$c = \pm \frac{15}{\sqrt{7}}$$

Hence, equation of common tangent is

$$y = 3\sqrt{\frac{2}{7}}x + \frac{15}{\sqrt{7}}$$

Q13 (2) - Tangent and its properties

Let $P(h, k)$ be the point of intersection of tangents $\Rightarrow y = mx \pm \sqrt{a^2 m^2 - b^2}$ passes through (h, k)

$$\Rightarrow k - mh = \pm \sqrt{a^2 m^2 - b^2}$$

$$\Rightarrow (k - mh)^2 = a^2 m^2 - b^2$$

$$\Rightarrow k^2 + m^2 h^2 - 2khm = a^2 m^2 - b^2$$

$$\Rightarrow m^2 (h^2 - a^2) - 2khm + (k^2 + b^2) = 0$$

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Given, $m_1 m_2 = c^2$

$$\Rightarrow \frac{k^2 + b^2}{n^2 - a^2} = c^2$$

$$\Rightarrow k^2 + b^2 = c^2 (h^2 - a^2)$$

$\therefore$ Locus of (h, k) is

$$y^2 + b^2 = c^2 (x^2 - a^2)$$

Q14 (3) - Tangent and its properties

Let $P(h, k)$ be the point of intersection of tangents. $\Rightarrow y = mx \pm \sqrt{a^2 m^2 - b^2}$ passes through (h, k)

$$\Rightarrow (k - mh)^2 = a^2 m^2 + b^2$$

$$\Rightarrow m^2 (h^2 - a^2) - 2khm + (k^2 + b^2) = 0$$

Given, $m_1 + m_2 = c^2$

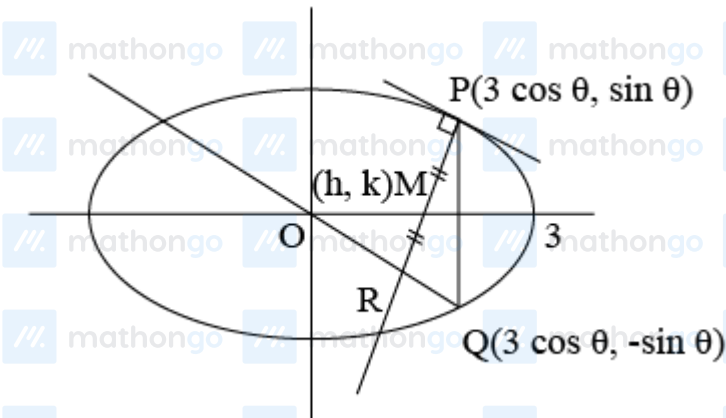
$$\Rightarrow \frac{2hk}{h^2 - a^2} = c^2$$

$$\Rightarrow 2xy = c^2 (x^2 - a^2)$$

Q15 (3) - Normal and its properties

Here, equation of normal at P , is

$$3x \sec \theta - y \operatorname{cosec} \theta = 8 \dots (i)$$



Equation of diameter OQ , is

$$x \sin \theta + 3y \cos \theta = 0 \dots (ii)$$

R is the point of intersection of (i) & (ii),

$$\therefore R \left(\frac{12}{5} \cos \theta, -\frac{4}{5} \sin \theta \right)$$

as $M(h, k)$ is the mid-point

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$$\Rightarrow h = \frac{3 \cos \theta + \frac{12}{5} \cos \theta}{2}, k = \frac{\sin \theta - \frac{4}{5} \sin \theta}{2}$$

$$\therefore \frac{10}{27}h = \cos \theta \text{ and } 10k = \sin \theta$$

Squaring and adding, we get

$$\left(\frac{10h}{27}\right)^2 + \frac{(10k)^2}{1} = 1$$

$$\text{or } \frac{100x^2}{(27)^2} + 100y^2 = 1$$

$\therefore$ locus is ellipse

Q16 (1) - Normal and its properties

Equation of tangent to the ellipse will be

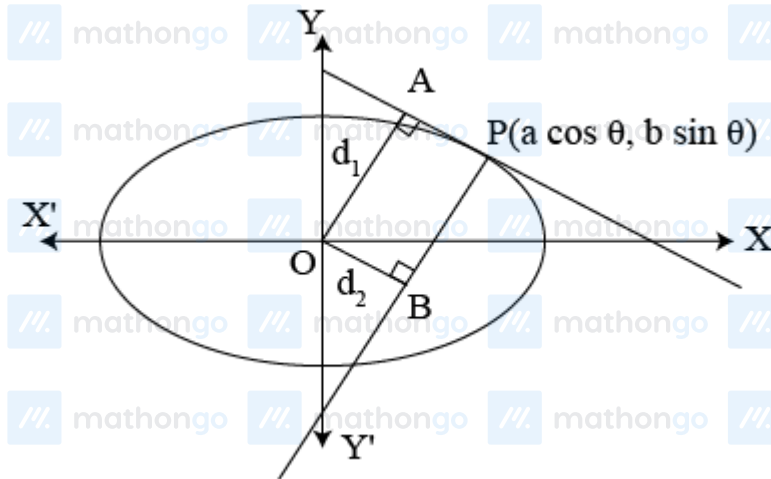
$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

$$d_1 = \left| \frac{ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \right| \dots (i)$$

Equation of normal to the ellipse will be

$$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$$

$$\therefore d_2 = \left| \frac{(a^2 - b^2) \sin \theta \cos \theta}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}} \right| \dots (ii)$$



$\therefore$ Area of rectangle $OAPB$ will be given as

$$A = d_1 d_2 = \frac{ab(a^2 - b^2)}{2\left(\frac{a^2}{2} + \frac{b^2}{2}\right)}$$

$$\text{where } \theta = \frac{\pi}{4}$$

$$= \frac{ab(a^2 - b^2)}{a^2 + b^2}$$

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Q17 (2) - Normal and its properties

The coordinates of extremity of the latusrectum which lies in the first quadrant are $\left(ae, \frac{b^2}{a}\right)$.

The equation of normal at (x_1, y_1) is

$$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = (a^2 - b^2)$$

Therefore, the equation of normal at $\left(ae, \frac{b^2}{a}\right)$

$$\frac{a^2x}{ae} - \frac{b^2y}{b^2/a} = a^2 - b^2$$

$$\Rightarrow ax - aey = e(a^2 - b^2)$$

$$\Rightarrow ax - aey = ea^2e^2$$

$$\Rightarrow x - ey = ae^3$$

This passes through the extremity of the minor axis, i.e. $(0, -b)$.

$$\therefore 0 + eb - ae^3 = 0 \Rightarrow b = ae^2$$

$$\Rightarrow b^2 = a^2e^4$$

$$\Rightarrow a^2(1 - e^2) = a^2e^4$$

$$\Rightarrow e^4 + e^2 - 1 = 0$$

Q18 (2) - Normal and its properties

Equation of normal to the ellipse at P is

$$ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2 \dots(i)$$

Equation of normal to the circle $x^2 + y^2 = a^2$ at point θ is

$$y = x \tan \theta \dots(ii)$$

Substituting the value of x in Eq. (i), we get $ax \sec \theta - bx \tan \theta \operatorname{cosec} \theta = a^2 - b^2$

$$(a - b)x \sec \theta = a^2 - b^2$$

$$\frac{(a-b)}{a^2-b^2}x = \cos \theta$$

$$\Rightarrow x = \frac{(a^2-b^2)}{a-b} \cos \theta = (a+b) \cos \theta$$

$$\Rightarrow y = (a+b) \sin \theta$$

Hence, required locus is $x^2 + y^2 = (a+b)^2$.

Q19 (4) - Normal and its properties

The equation of normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at $P(a \sec \theta, b \tan \theta)$ is

$$ax \cos \theta + by \cot \theta = a^2 + b^2 \dots(i)$$

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Similarly, the equation at $Q(a \sec \phi, b \tan \phi)$ is

$$ax \cos \phi + by \cot \phi = a^2 + b^2 \dots (ii)$$

We have, $\theta + \phi = \frac{\pi}{2}$

$$ax \cos\left(\frac{\pi}{2} - \theta\right) + by \cot\left(\frac{\pi}{2} - \theta\right) = a^2 + b^2$$

$$ax \sin \theta + by \tan \theta = a^2 + b^2 \dots (iii)$$

Now, multiplying in Eq. (i) by $\sin \theta$ and Eqs. (iii) by $\cos \theta$ and then subtracting, we get

$$by(\cos \theta - \sin \theta) = (a^2 + b^2)(\sin \theta - \cos \theta)$$

$$\therefore y = k = -\left(\frac{a^2 + b^2}{b}\right)$$

Q20 (2) - Normal and its properties

Equation normals of slope m to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$y = mx \pm \frac{m(a^2 + b^2)}{\sqrt{a^2 - b^2 m^2}}$$

which is biquadratic equation in m .

$\therefore$ 4 normals can be drawn from an external point of a hyperbola.

Q21 (3) - Normal and its properties

Let point on the hyperbola $x^2 - y^2 = 4$ be $(2 \sec \theta, 2 \tan \theta)$.

Equation of tangent at $(2 \sec \theta, 2 \tan \theta)$

$$\frac{x}{2} \sec \theta - \frac{y}{2} \tan \theta = 1$$

x -intercept, i.e.

$$a_1 = 2 \cos \theta$$

Y -intercept, i.e. $b_1 = -2 \cot \theta$

Now, the equation of normal at $(2 \sec \theta, 2 \tan \theta)$ is

$$2x \cos \theta + 2y \cot \theta = a^2 + b^2$$

$$x\text{-intercept, i.e. } a_2 = \frac{a^2 + b^2}{2 \cos \theta}$$

$$y\text{-intercept, i.e. } b_2 = \frac{a^2 + b^2}{2 \cot \theta}$$

$$\therefore a_1 a_2 + b_1 b_2 = 2 \cos \theta \times \frac{a^2 + b^2}{2 \cos \theta} + (-2 \cot \theta) \left(\frac{a^2 + b^2}{2 \cot \theta} \right)$$

$$= a^2 + b^2 - a^2 - b^2 = 0$$

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Q22 (3) - Normal and its properties

Let the normal points be (x_1, y_1) , (x_2, y_2) , (x_3, y_3) and (x_4, y_4) . Equation of normal is

$$\frac{a^2(x-x_i)}{x_i} = \frac{b^2(y-y_i)}{y_i} \quad [\text{where } i = 1, 2, 3, 4]$$

Let these normal passes through the point (h, k)

$$\text{So, } \frac{a^2(h-x)}{x} = \frac{b^2(k-y)}{y}$$

$$\Rightarrow a^2y(h-x) = b^2x(k-y)$$

$$\text{i.e. } a^2y(x-h) + b^2x(y-k) = 0$$

Q23 (2) - Chord of Contact, Diameter and Conjugate Diameter

Let $P(h, k)$ be the given point. Then, the chord of contact of tangents drawn from P to the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is}$$

$$\frac{hx}{a^2} + \frac{ky}{b^2} = 1 \dots(i)$$

This subtends a right angle at the centre $C(0, 0)$ of the ellipse. The combined equation of the pair of straight lines joining C to the points of intersection of Eq. (i) and the ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \left(\frac{hx}{a^2} + \frac{ky}{b^2} \right)^2$$

this equation represents a pair of perpendicular straight lines $\therefore \frac{1}{a^2} - \frac{h^2}{a^4} + \frac{1}{b^2} - \frac{k^2}{b^4} = 0$

$$\Rightarrow \frac{h^2}{a^4} + \frac{k^2}{b^4} = \frac{1}{a^2} + \frac{1}{b^2}$$

Q24 (2) - Chord of Contact, Diameter and Conjugate Diameter

The equation of ellipse is

$$\frac{x^2}{10} + \frac{y^2}{6} = 1$$

$$\text{Or } 3x^2 + 5y^2 = 30 \dots(i)$$

Let (h, k) be the mid-point of chord intercepted on the line $2x - y + 3 = 0$ by the ellipse (i). Then, the equation of chord is

$$T = S'$$

$$\text{i.e. } 3hx + 5ky - 30 = 3h^2 + 5k^2 - 30$$

$$\text{or } 3hx + 5ky = 3h^2 + 5k^2 \dots(ii)$$

$$\text{The equation of line is } 2x - y + 3 = 0 \dots(iii)$$

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Clearly, Eqs. (ii) and (iii) represent the same lines.

$$\Rightarrow h = \frac{2\lambda}{3}, k = \frac{-\lambda}{5} \text{ and } 3h^2 + 5k^2 = -3\lambda$$

$$\Rightarrow 3 \left(\frac{4\lambda^2}{9} \right) + 5 \left(\frac{\lambda^2}{25} \right) = -3\lambda$$

$$\Rightarrow \frac{4\lambda^2}{3} + \frac{\lambda^2}{5} = -3\lambda$$

$$\Rightarrow 23\lambda^2 = -45\lambda$$

$$\Rightarrow \lambda = 0, \lambda = -\frac{45}{23}$$

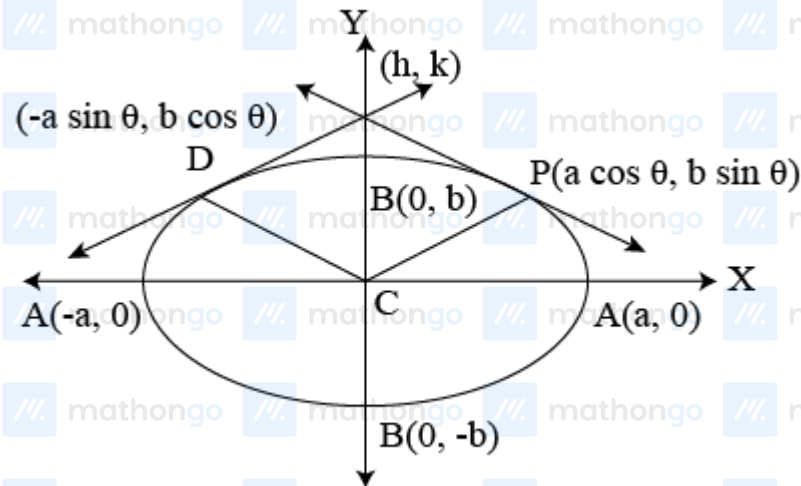
But $\lambda = 0$ is not possible.

$$\text{So, } \lambda = -\frac{45}{23}. \text{ Now, } h = \frac{2\lambda}{3}, k = -\frac{\lambda}{5} \Rightarrow h = -\frac{30}{23}, k = \frac{9}{23}$$

Hence, the required coordinates of the mid-point are $\left(-\frac{30}{23}, \frac{9}{23}\right)$

Q25 (3) - Chord of Contact, Diameter and Conjugate Diameter

Let CP and CD be two conjugate semi-diameters of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Then, the eccentric angles of P and D are θ and $\frac{\pi}{2} + \theta$, respectively. So, the coordinates of P and D are $(a \cos \theta, b \sin \theta)$ and $(-a \sin \theta, b \cos \theta)$ respectively.



Equations of the tangents at P and D are

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1 \dots (i)$$

$$\text{and } -\frac{x}{a} \sin \theta + \frac{y}{b} \cos \theta = 1 \dots (ii)$$

Let (h, k) be the point of intersection of Eqs. (i) and (ii). Then,

$$\frac{h}{a} \cos \theta + \frac{k}{b} \sin \theta = 1$$

$$\text{and } \frac{-h}{a} \sin \theta + \frac{k}{b} \cos \theta = 1$$

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$$\Rightarrow \left(\frac{h}{a} \cos \theta + \frac{k}{b} \sin \theta \right)^2 + \left(-\frac{h}{a} \sin \theta + \frac{k}{b} \cos \theta \right)^2 = 1 + 1$$

$$\Rightarrow \frac{h^2}{a^2} + \frac{k^2}{b^2} = 2$$

Hence, the locus of (h, k) is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$, which represents an ellipse.

Q26 (1) - Chord of Contact, Diameter and Conjugate Diameter

Let (h, k) be the mid-point of PD . Then,

$$2h = a \cos \theta - a \sin \theta$$

$$\text{and } 2k = b \sin \theta + b \cos \theta$$

$$\Rightarrow \frac{2h}{a} = \cos \theta - \sin \theta$$

$$\text{and } \frac{2k}{b} = \sin \theta + \cos \theta$$

$$\Rightarrow \frac{4h^2}{a^2} + \frac{4k^2}{b^2} = (\cos \theta - \sin \theta)^2 + (\sin \theta + \cos \theta)^2$$

$$\Rightarrow \frac{h^2}{a^2} + \frac{k^2}{b^2} = \frac{1}{2}$$

Hence, the locus of (h, k) is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{2}$.

Q27 (3) - Chord of Contact, Diameter and Conjugate Diameter

Let $P(a \cos \theta, b \sin \theta)$ and $D(-a \sin \theta, b \cos \theta)$ be the

extremities of the conjugate semi-diameters CP and CD of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The equations of the circles with CP and CQ as diameters are

$$x(x - a \cos \theta) + y(y - b \sin \theta) = 0$$

$$\text{and } x(x + a \sin \theta) + y(y - b \cos \theta) = 0$$

$$\Rightarrow x^2 + y^2 = ax \cos \theta + by \sin \theta \dots (i)$$

$$\text{and } x^2 + y^2 = -ax \sin \theta + by \cos \theta \dots (ii)$$

Let $R(h, k)$ be the intersection of Eqs. (i) and (ii), then

$$h^2 + k^2 = ah \cos \theta + bk \sin \theta$$

$$\text{and } h^2 + k^2 = -ah \sin \theta + bk \cos \theta$$

Q28 (1) - Asymptotes and Rectangular Hyperbola

Since, $\frac{x}{a} \pm \frac{y}{b}$ are the two asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Now, $m_1 = -\frac{b}{a}$ and $m_2 = \frac{b}{a}$

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$$\therefore \tan \theta = \frac{\frac{b}{a} + \frac{b}{a}}{1 - \frac{b}{a} \times \frac{b}{a}} = \frac{2ab}{a^2 - b^2}$$

$$\Rightarrow \theta = \tan^{-1} \frac{2ab}{a^2 - b^2} = 2 \tan^{-1} \frac{b}{a}$$

Q29 (2) - Asymptotes and Rectangular Hyperbola

We have, hyperbola $x^2 - 2y^2 - 2 = 0$

$$\Rightarrow \frac{x^2}{2} - y^2 = 1$$

Let any point on it is $(\sqrt{2} \sec \theta, \tan \theta)$ and its asymptotes are

$$\frac{x}{\sqrt{2}} + y = 0 \text{ and } \frac{x}{\sqrt{2}} - y = 0$$

Let p_1 and p_2 be the length of perpendicular from $(\sqrt{2} \sec \theta, \tan \theta)$ to the asymptotes $\frac{x}{\sqrt{2}} + y = 0$ and

$$\frac{x}{\sqrt{2}} - y = 0, \text{ respectively.}$$

$$\text{Now, } p_1 = \frac{\sec \theta + \tan \theta}{\sqrt{\frac{3}{2}}} \text{ and } p_2 = \frac{\sec \theta - \tan \theta}{\sqrt{\frac{3}{2}}}$$

$$\text{Hence, } p_1 \cdot p_2 = \frac{\sec^2 \theta - \tan^2 \theta}{\frac{3}{2}} = \frac{2}{3}$$

Q30 (2) - Asymptotes and Rectangular Hyperbola

The equation of hyperbola having asymptotes are

$$3x - 4y + 7 = 0 \text{ and } 4x + 3y + 1 = 0 \text{ is}$$

$$(3x - 4y + 7)(4x + 3y + 1) + \lambda = 0$$

Since, their hyperbola passes through the origin,

$$\text{i.e. } 7 + \lambda = 0 \Rightarrow \lambda = -7$$

$\therefore$ Equation of hyperbola is

$$(3x - 4y + 7)(4x + 3y + 1) - 7 = 0$$

Q31 (3) - Asymptotes and Rectangular Hyperbola

Asymptotes are parallel to the lines

$$2x + 3y = 0 \text{ and } 3x + 2y = 0$$

$\therefore$ Equation of asymptotes are

$$2x + 3y = \lambda \text{ and } 3x + 2y = \mu$$

Since, this passes through the centre of the hyperbola (1,2)

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$$\lambda = +8 \text{ and } \mu = 7$$

Thus, the equation of asymptotes are

$$2x + 3y - 8 = 0 \text{ and } 3x + 2y - 7 = 0$$

Let the equation of hyperbola be

$$(2x + 3y - 8)(3x + 2y - 7) + k = 0$$

Since, this passes through (5,3) Hence,

$$(10 + 9 - 8)(15 + 6 - 7) + k = 0$$

$$(11)(14) + k = 0$$

$$k = -154$$

$$\therefore (2x + 3y - 8)(3x + 2y - 7) - 154 = 0$$

$$\Rightarrow \lambda = 154$$

Q32 (4) - Asymptotes and Rectangular Hyperbola

Let the equation of circle be

$$x^2 + y^2 + 2gx + 2fy + k = 0 \dots(i)$$

$$\text{We have, radius} = g^2 + f^2 - k = r^2$$

Let $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$ and $S(x_4, y_4)$ be the point of intersection of Eq. (1) and $xy = c^2$.

$$x^2 + \frac{c^4}{x^2} + 2gx + \frac{2fc^2}{x} + k = 0$$

$$\Rightarrow x^4 + 2gx^3 + kx^2 + 2fc^2x + c^4 = 0$$

$$\Rightarrow x_1 + x_2 + x_3 + x_4 = -2g$$

$$\text{and } \Sigma x_1x_2 = k$$

$$\therefore x_1^2 + x_2^2 + x_3^2 + x_4^2 = (x_1 + x_2 + x_3 + x_4)^2 - 2\Sigma x_1x_2$$

$$= 4g^2 - 2k$$

$$\text{Similarly, } y_1^2 + y_2^2 + y_3^2 + y_4^2 = 4f^2 - 2k$$

$$\therefore CP^2 + CQ^2 + CR^2 + CS^2$$

$$= x_1^2 + x_2^2 + x_3^2 + x_4^2 + y_1^2 + y_2^2 + y_3^2 + y_4^2$$

$$= 4g^2 - 2k + 4f^2 - 2k$$

$$= 4(g^2 + f^2 - k) = 4r^2$$

Q33 (3) - Asymptotes and Rectangular Hyperbola

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We have, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$e^2 = 1 + \frac{b^2}{a^2}$$

and asymmetrical are $\frac{x}{a} \pm \frac{y}{b} = 0$

Angle between asymptotes are

$$\theta = 2 \tan^{-1} \frac{b}{a} \Rightarrow \tan \frac{\theta}{2} = \frac{b}{a}$$

$$\therefore \sec \frac{\theta}{2} = \frac{\sqrt{a^2 + b^2}}{a} = \sqrt{1 + \frac{b^2}{a^2}} = e$$

Q34 (3) - Asymptotes and Rectangular Hyperbola

The equation of normal at $(ct, \frac{c}{t})$ to the hyperbola $xy = c^2$ is

$$xt^3 - yt - ct^4 + c = 0 \dots(i)$$

Since, Eq. (i) intersect at hyperbola $(ct_1, \frac{c}{t_1})$.

$$ct_1 t^3 - \frac{c}{t_1} t - ct^4 + c = 0$$

$$\Rightarrow t^3 t_1 = \frac{t}{t_1} + t^4 - 1$$

$$\Rightarrow t^3 t_1^2 - t - t^4 t_1 + t_1 = 0$$

$$\Rightarrow (t^3 t_1^2 - t^4 t_1) + (t_1 - t) = 0$$

$$\Rightarrow t^3 t_1 (t_1 - t) + 1 (t_1 - t) = 0$$

$$\Rightarrow (t_1 - t) (t^3 t_1 + 1) = 0$$

$$\Rightarrow t_1 \neq t$$

$$\text{Ci } t^3 t_1 + 1 = 0$$

$$\therefore t^3 t_1 = -1$$

Q35 (4) - Asymptotes and Rectangular Hyperbola

We have, $(x, y_i), i = 1, 2, 3, 4$ are the points on the rectangular hyperbola $xy = c^2$ Equation of normal to the hyperbola $xy = c^2$ at point $(ct, \frac{c}{t})$ is

$$ct^4 - t^3 x + ty - c = 0$$

It passes through (h, k) , then

$$ct^4 - t^3 h + tk - c = 0$$

Which is a biquadratic in t

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Let its roots be t_1, t_2, t_3 and t_4 , then

$$\Sigma t_1 = \frac{h}{c}, \Sigma t_1 t_2 = 0$$

$$\Sigma t_1 t_2 t_3 = -\frac{k}{c}$$

$$\text{and } t_1 t_2 t_3 t_4 = -1$$

$$\text{Now, } \Sigma x_i = c \Sigma t_1 = c \cdot \frac{h}{c} = h$$

$$\text{and } \Sigma y_i = c \left(\Sigma \frac{1}{t_1} \right)$$

$$= c \left(\frac{\Sigma t_1 t_2 t_3}{t_1 t_2 t_3 t_4} \right)$$

$$= c \cdot \left(-\frac{k}{c} \right) = -k$$

$$= -\frac{(x_1 + x_2 + x_3 + x_4)(y_1 + y_2 + y_3 + y_4)}{x_1 x_2 x_3 x_4}$$

$$= -\frac{hk}{-c^4} \quad [x_1 x_2 x_3 x_4 = c^4 t_1 t_2 t_3 t_4 = -c^4]$$

$$= -\frac{hk}{c^4}$$